

User Manual

ESP32 Temperature, Humidity, and Gas Monitoring System

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Introduction

This project is an ESP32-based monitoring system that measures temperature, humidity, and gas levels using connected sensors. Readings can be viewed through a local web dashboard and saved for later review.

Purpose

- Monitor environmental conditions in real time
- View readings on a local website using the ESP32 device IP address
- Save session data for later checking
- Provide a simple academic prototype for monitoring and logging

Benefits

- Real-time monitoring of temperature, humidity, and gas
- Browser-based access on devices connected to the same WiFi
- Saved readings for later review
- Simple setup for school project demonstration

Getting Started

Before using the device, prepare:

- Laptop or computer
- Safe USB power source for the ESP32
- WiFi network
- Google account if you want to save data using Google Sheets

Important notes

- ESP32 and viewing device must be on the same WiFi network
- Use the system in a dry, ventilated area
- This system is for educational and prototype use

Device Setup and Access

- Connect the ESP32 device to a laptop using USB
- Power the device using a safe USB power source
- Make sure the ESP32 is on the configured WiFi network
- Connect the laptop or phone to the same WiFi
- Open Serial Monitor and wait for the local IP address
- Open the IP address in a browser to access the monitoring website

Optional Google Sheets setup

- Use a Google account
- Open the backend setup in D:\repos\GasSensor-ESP32\GoogleSheets-Backend
- Deploy the Apps Script backend as a Web App
- Put the generated Web App URL in the ESP32 project settings

Repository reference

<https://github.com/tychesama/GasSensor-ESP32>

Using the Website

Main functions

- View live temperature, humidity, and gas readings
- Review saved session data
- Change the display between horizontal and vertical layouts
- Check session-based history from the website
- Download or review saved data if available

Usage flow

- Power the device
- Wait for WiFi connection
- Open the IP address shown in Serial Monitor
- Use the website to monitor current readings
- Open saved data or history when needed

Troubleshooting

- No power, check the USB cable and power source
- No IP address, check WiFi settings and restart the device
- Website not opening, make sure both devices are on the same WiFi network
- No sensor readings, check wiring and sensor connections
- Saved data not working, recheck the Google Sheets backend setup

Maintenance

- Keep it away from dust and water
- Keep ventilation areas clear
- Do not expose it to rough handling
- Keep it away from children
- Check wires and sensor connections if readings become unstable

Warranty and Support

This project is an academic prototype and has no formal commercial warranty.

Support contacts:

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